

Plasmon-polariton modes guided by a metal film of finite width bounded by different dielectrics

Pierre Berini

*School of Information Technology and Engineering, University of Ottawa,
Ottawa, Ontario, K1N 6N5, Canada
berini@site.uottawa.ca*

Abstract: The properties of some purely bound plasmon-polariton modes guided by an asymmetric waveguide structure composed of a thin lossy metal film of finite width supported by a dielectric substrate and covered by a different dielectric superstrate are presented for what is believed to be the first time. The mode spectrum supported by these structures is quite different from the spectrum supported by corresponding asymmetric slab structures or similar finite-width symmetric waveguides. Unlike these limiting cases, the dispersion with film thickness exhibits an unusual oscillatory character that is explained by a “switching” of constituent interface modes. This mode switching is unique to asymmetric finite-width structures. Above a certain cut-off film thickness, the structure can support a long-ranging mode and its attenuation decreases very rapidly with decreasing film thickness, more so than the long-ranging mode in symmetric structures. Also, the cutoff thickness of the long-ranging mode is larger than the cutoff thickness of the long-ranging mode in the corresponding asymmetric slab waveguide, which implies that propagation along finite-width films is more sensitive to the asymmetry in the structure than propagation along a similar slab structure. Both of these results are potentially useful for the transmission and control of optical radiation.

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OCIS codes: (240.6680) Surface plasmons; (130.2790) Guided waves

References and Links

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1. Introduction

It is well-known that an interface between semi-infinite materials having positive and negative dielectric constants can guide electromagnetic surface waves. In the case of a metal-dielectric interface, where the metal has an apparent negative dielectric constant at the optical free-space wavelength of interest, the waves are termed plasmon-polariton waves and propagate as electromagnetic fields coupled to surface plasmons (surface plasma oscillations) comprised of conduction electrons in the metal. A metal film of a certain thickness bounded by dielectrics above and below can serve as an optical slab (infinitely wide) waveguide. When the film is thin enough, the plasmon-polariton modes guided by the top and bottom interfaces become coupled due to field tunneling through the metal, thus creating supermodes that exhibit dispersion with metal thickness. The modes supported by infinitely wide symmetric and

asymmetric metal film structures are well-known; some notable published works include references [1-3].

Metal films of finite thickness and finite width offer 2-D field confinement in the plane transverse to the direction of propagation. These waveguides may prove to be quite useful since they can be used in principle for optical signal transmission and routing over short distances or to construct passive components such as couplers and power splitters. The purely bound mode spectrum guided by symmetric structures comprised of a thin metal film of finite width embedded in a homogeneous dielectric has recently been investigated theoretically [4], and results suggest that power attenuation values near 0.1 dB/cm are achievable with even lower values possible. The operation of straight waveguide segments has recently been demonstrated in an end-fire experiment [5]. The results obtained so far suggest that finite-width structures may find widespread use as an integrated optics technology. Further research work along this direction is thus motivated.

Asymmetric structures, comprised of a metal film of finite width bounded by different substrate and superstrate dielectrics, form an important class of waveguides, yet such structures had hitherto not been studied either theoretically or experimentally. This paper reports the first set of results describing some of the waveguiding characteristics of these structures. Their operation differs significantly from the operation of corresponding asymmetric slab waveguides or similar symmetric finite-width structures.

2. Mode Dispersion and Evolution With Film Thickness

The structure of interest is shown as the inset in Figure 1 (a). It consists of a metal film of thickness t , width w and equivalent permittivity ϵ_2 , supported by a semi-infinite homogeneous dielectric substrate of permittivity ϵ_1 and covered by a semi-infinite homogeneous dielectric superstrate of permittivity ϵ_3 . The Cartesian coordinate axes used for the analysis are also shown. The structure is invariant along the propagation axis (z axis, out of the page). The mode fields vary along this dimension according to $e^{-\gamma z}$ where $\gamma = \alpha + j\beta$ is the complex propagation constant of the mode, α being its attenuation constant and β its phase constant. Material parameters taken from the standard work on slab structures [2] were used in order to facilitate comparisons: the optical free space wavelength of analysis is set to $\lambda_0 = 0.633 \mu\text{m}$, the metal film is silver, which has at the wavelength of interest a relative permittivity of $\epsilon_{r,2} = -19 - j0.53$, and the relative permittivity of the bottom and top dielectric regions are set to $\epsilon_{r,1} = 4$ and $\epsilon_{r,3} = 3.61$, respectively.

The modes supported by the structure are obtained by solving a suitably defined boundary value problem based on Maxwell's equations written in the frequency domain. The boundary value problem cannot be solved analytically so it is solved numerically by applying the Method of Lines (MoL) which is a well-known numerical technique. The application of the MoL to various electromagnetic problems is well established and its use for modeling waveguide structures such as those of concern in this paper is summarized in [4]. The propagation constant and the spatial distribution over the waveguide cross-section of the six field components (and the associated Poynting vector) related to a mode are computed by the MoL.

The study consists of the analysis of the structure shown as the inset in Figure 1 (a) for the case $w = 1 \mu\text{m}$ and the material parameters given above. The dispersion with film thickness of the first 5 purely bound modes has been computed using the MoL and the resulting curves are shown in Figure 1. The dispersion curves of the a_b and s_b modes guided by the corresponding slab waveguide ($w = \infty$) are given for comparison. The dispersion curve of the long-ranging ss_b^0 mode guided by a corresponding symmetric structure ($w = 1 \mu\text{m}$, $\epsilon_{r,1} = \epsilon_{r,3} = 4$) is also shown as the curve labeled ss_b^0 ,sym.

In asymmetric structures of the type considered in this paper, true field symmetry exists only with respect to the y axis. With respect to the horizontal dimension, the modes have a symmetric-like or asymmetric-like field distribution with field localization along either the bottom or top metal-dielectric interface. As in slab structures, the modes that have a symmetric-like distribution with respect to the horizontal dimension are localized along the metal-dielectric interface with the lowest dielectric constant, while modes that have an asymmetric-like distribution with respect to this axis are localized along the metal-dielectric interface with the highest dielectric constant.

The modes supported by finite-width structures are not TM in nature but for films where $w/t > 1$, the E_y field component dominates. The nomenclature discussed in [4] to describe the modes supported by symmetric structures can be used to describe the modes guided by asymmetric structures. However, care must be taken to identify the modes when the metal film is optically thick, before significant field coupling begins to occur through the film and while the origin of the mode can be identified unambiguously. The number of extrema in the E_y field component of the mode is counted along the lateral dimension at the metal-dielectric interface where the field is *localized* and this number is then used in the nomenclature.

The sa_b^0 , aa_b^0 , ss_b^0 and as_b^0 modes are the fundamental modes guided by the structure shown as the inset of Figure 1 (a). The sa_b^0 and aa_b^0 modes are comprised of coupled corner modes, resembling the corresponding modes in a similar symmetric structure, except that the fields are localized at the substrate-metal interface. These two modes do not change in character as the thickness of the film decreases and a reduction in film width would eventually break the degeneracy shown in Figure 1. The behavior of these modes as the thickness of the film is reduced is consistent with their behavior in symmetric structures.

The ss_b^0 and as_b^0 modes change substantially in character with decreasing film thickness. Figure 2 shows the spatial distribution of the E_y field component related to these modes for two film thicknesses: $t=100$ nm and $t=60$ nm. When the film is optically thick, as in parts (a) and (c), the modes show their defining character and are identified unambiguously. Parts (a) and (b) illustrate how the ss_b^0 mode evolves from a symmetric-like mode having fields localized at the metal-superstrate interface to an asymmetric-like mode having fields localized along the substrate-metal interface as the film thickness is decreased to 60 nm. Parts (c) and (d) show a similar evolution for the as_b^0 mode. This change in character is also apparent in their dispersion curves: the modes follow the general behavior of a symmetric-like mode for optically large thicknesses but then slowly change to follow the behavior of an asymmetric-like mode as the thickness decreases. Since the substrate dielectric constant is larger than the superstrate dielectric constant, the mode fields are “pulled” from the superstrate interface to the substrate interface as the metal film becomes thinner. The behavior of these modes as film thickness decreases is unexpected and differs completely from their behavior in a symmetric structure. In a symmetric structure both modes exhibit a *decreasing* attenuation with decreasing film thickness, the ss_b^0 mode is the main long-ranging mode, and the as_b^0 mode has a cutoff thickness below which propagation ceases [4]. In this asymmetric structure, both modes exhibit an *increasing* attenuation with decreasing film thickness, and the as_b^0 mode does not exhibit a cutoff thickness.

It was noted in [4] that the modes supported by a metal film of finite width are in fact supermodes created from a coupling of “edge” and “corner” modes supported by each metal-dielectric interface defining the structure. As the dimensions of the metal film change, the level of coupling between the interface modes changes, leading to dispersion and possibly a change in character of the supermode. The results of the current study further support this view. In asymmetric structures, the modes are also supermodes created in a similar manner, except that *dissimilar* interface modes may couple to each other to create the supermode. For instance, a mode having one field extremum along the top interface (along the metal-superstrate edge between the corners) may couple with a mode having three field extrema along the bottom interface. The selection criteria dictating which interface modes will couple

to form a supermode are: a sharing of field symmetry with respect to the vertical axis and a similarity in the value of their propagation constants. As the film dimensions change, the interface modes selected can also change, leading to unexpected behavior in the supermode. This is in contrast to symmetric structures, where identical interface modes only will couple to form the supermode, and in this case, mode behavior with changing film dimensions is easily predictable.

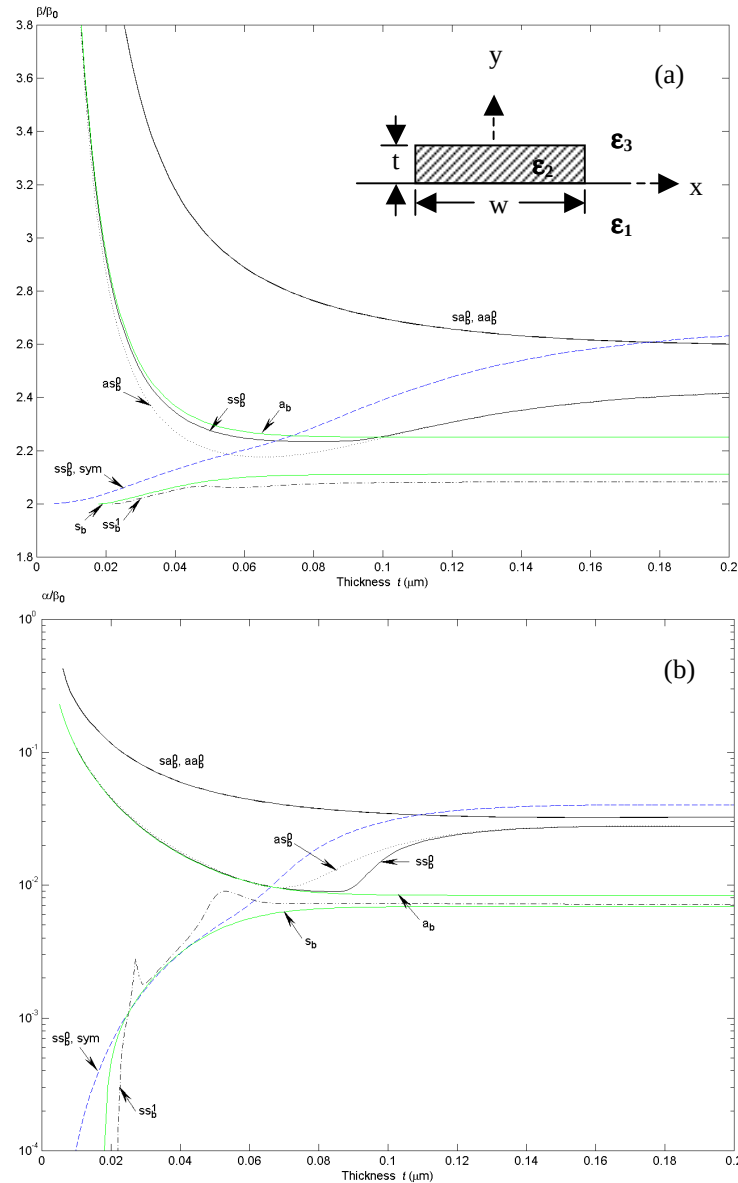


Fig. 1. Dispersion characteristics of some purely bound modes supported by an asymmetric metal film waveguide of width $w = 1 \mu\text{m}$. (a) Normalized phase constant. (b) Normalized attenuation constant.

3. Long-ranging mode

The only potentially long-ranging mode found for this structure, at the wavelength of analysis, is the ss_b^1 mode. As shown in Figure 1 (b), the mode has a cutoff thickness near $t = 22 \text{ nm}$ and

its attenuation drops very quickly with decreasing thickness above this value. It is also clear from the figure that the rate of decrease in attenuation is much greater for this mode near cut-off than the rate of decrease in attenuation related to the ss_b^0 mode guided by the corresponding symmetric structure (curve labeled ss_b^0, sym). Thus, a means for range extension similar to that found in slab structures [6], also exists for metal films of finite width.

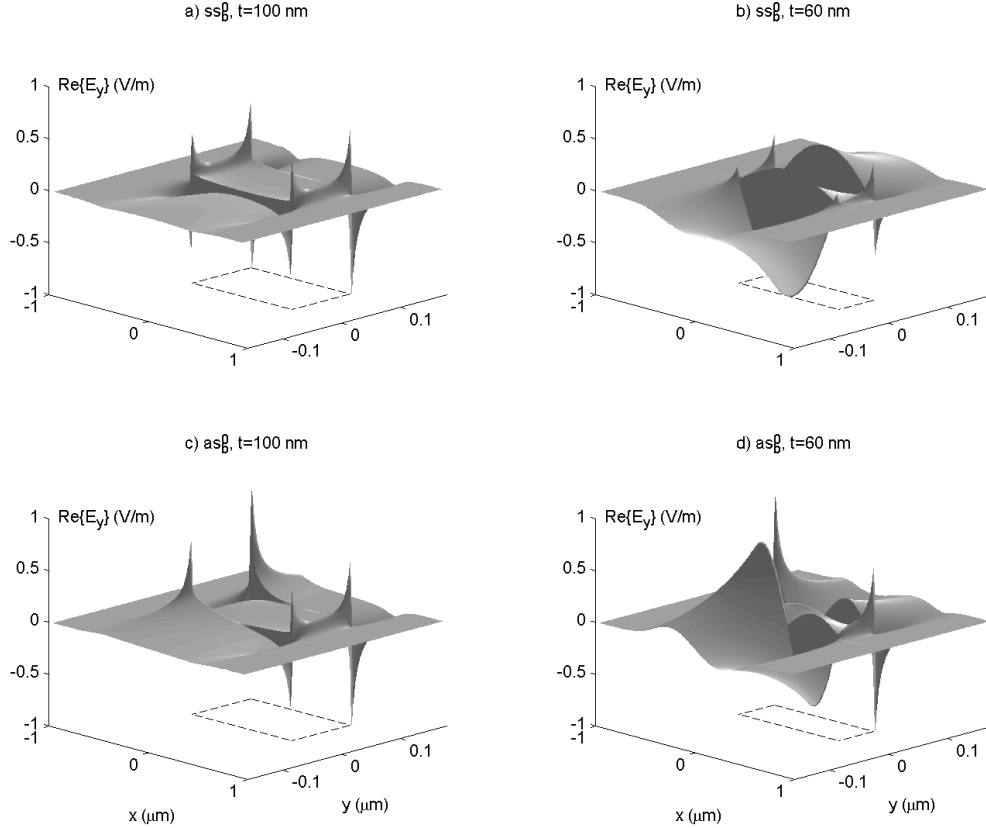


Fig. 2. Spatial distribution of the E_y field component related to the ss_b^0 and as_b^0 modes for two film thicknesses. The waveguide cross-section is located in the x - y plane and the metal region is outlined as the rectangular dashed contour. The field distributions are normalized such that $\max|\Re\{E_y\}|=1$.

Figure 3 shows contour plots of $\Re\{S_z\}$ associated with the ss_b^1 mode for four film thicknesses. S_z is the z -directed component of the Poynting vector and its spatial distribution is computed from the spatial distribution of the mode fields using $S_z = (\mathbf{E}_x \mathbf{H}_y^* - \mathbf{E}_y \mathbf{H}_x^*) / 2$ where $*$ denotes the complex conjugate. From this figure it is noted that the top and bottom interface modes comprising this supermode are indeed different from each other. From part (a), it is seen that the bottom interface mode has three extrema, and thus is of higher order than the top interface mode which has one extremum between the corners. In this structure, the substrate has a higher dielectric constant than the superstrate so the phase constant of a particular substrate-metal interface mode is higher than the phase constant of the same mode at the metal-superstrate interface. Higher-order modes have in general smaller values of phase constant compared to lower-order modes, so in structures having $\epsilon_3 < \epsilon_1$, all supermodes are comprised of a bottom interface mode of the same order or higher than the top interface mode. If $\epsilon_3 > \epsilon_1$ then the opposite situation occurs. As mentioned earlier the particular interface modes that are selected share a symmetry along the vertical axis and have similar propagation constants.

A careful inspection of the fields associated with the ss_b^1 mode at various film thicknesses reveals that as the thickness of the film decreases, the mode fields evolve in a smooth manner but in addition a change or “switch” of the constituent interface and/or corner modes also occurs. In all cases, such changes are consistent with the directions taken by the dispersion curve as the film thickness decreases, thus explaining the oscillations in the curve for this mode seen Figure 1. In particular, a change of the bottom edge mode occurs near $t = 27$ nm and results in a peak in the attenuation curve of the mode near this thickness as seen in Figure 1 (b). This change of the bottom interface mode is also apparent by comparing parts (c) and (d) of Figure 3.

Though the attenuation of the ss_b^1 mode drops quickly near its cut-off thickness, it should be remembered that the field confinement does so as well. Part (d) of Figure 3 corresponds to a film thickness slightly above cutoff, representative of a thickness that would be used to observe this long-ranging mode experimentally. Comparing parts (a) and (d) reveals that the mode fields do indeed extend further into the background dielectrics as the thickness of the film is reduced. Furthermore, it is observed from part (d) that near cutoff the fields have strong extrema along the top and bottom interfaces. These extrema may result in increased coupling losses in an end-fire configuration compared to the ss_b^0 mode of a symmetric structure since field distributions would not overlap as cleanly. Also, the fact that the mode would be used near its cutoff thickness implies that very tight tolerances are required in the fabrication of structures and that all metal-dielectric interfaces should be of the highest quality. Nevertheless, it should be possible to observe propagation of this mode in a suitable structure excited in an end-fire manner.

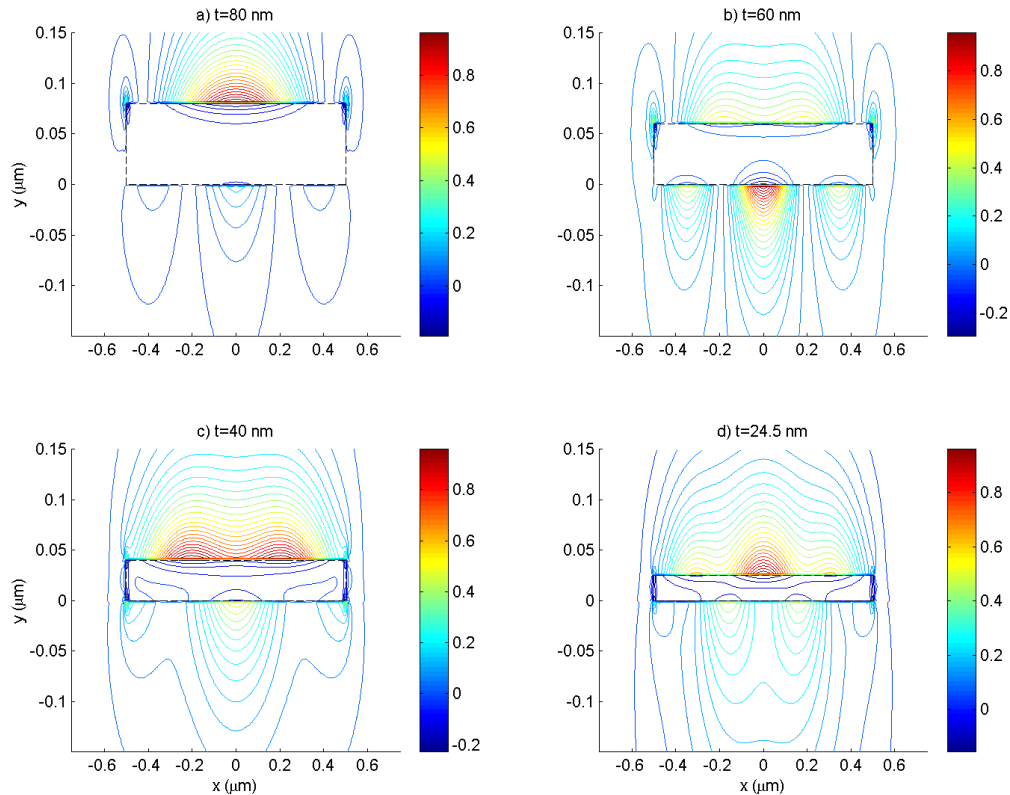


Fig. 3. Contour plot of $\Re\{S_z\}$ associated with the long-ranging ss_b^1 mode as a function of film thickness. In all cases, the outline of the metal film is shown as the rectangular dashed contour.

It is also noted from Figure 1 that the cutoff thickness of the ss_b^1 mode ($t \sim 22$ nm) is greater than the cutoff thickness of the s_b mode ($t \sim 18$ nm) supported by the corresponding slab structure. This implies that a long-ranging mode guided by a thin narrow metal film is more sensitive to differences in the superstrate and substrate permittivities than the s_b mode supported by the corresponding slab structure. This is reasonable in light of the fact that in finite-width structures the mode fields tunnel through the metal as in slab structures, but in addition the fields also wrap around the metal film. This enhanced sensitivity of long-ranging modes in thin metal films of finite width to structure asymmetry is potentially useful, in that a small induced change in substrate or superstrate refractive index has a greater impact on propagation compared to corresponding slab structures.

4. Conclusion

Some purely bound optical modes guided by a thin metal film of finite width supported by a semi-infinite dielectric and covered by a different semi-infinite dielectric have been characterized and described. The modes guided differ significantly from those supported by the corresponding asymmetric slab waveguide and a similar symmetric finite-width structure. In the structure analyzed, all four of the fundamental modes exhibit an increasing attenuation with decreasing film thickness, which implies that in an asymmetric finite-width structure the ss_b^0 mode may not be long-ranging. This result is unexpected since in a symmetric structure, the ss_b^0 and as_b^0 modes exhibit a decreasing attenuation with decreasing film thickness, and the ss_b^0 mode is the main long-ranging mode. A feature that is unique to finite-width asymmetric structures is that the guided modes may be composed of different interface modes. This leads to a rather complicated dispersion behavior as the film thickness is varied. A long-ranging mode has been found, exhibiting a rapidly diminishing attenuation near its cutoff thickness. The rate of decrease of its attenuation is greater than the rate related to the ss_b^0 mode in a corresponding symmetric structure. It also appears that the long-ranging modes guided by metal films of finite width are more sensitive to the asymmetry in the structure than the s_b mode guided by corresponding slab waveguides. Both of these results are potentially useful for the transmission and control of optical radiation over short distances.